Project Brief: Local Weather Tracker

Objective

Build a simple Python application that:

1. **Retrieves live weather data** from an external API (Open-Meteo).
2. **Stores** observations in a **PostgreSQL** database.
3. Exposes a **CRUD Web Service** (using FastAPI) to:
    - **Create** — ingest and save new observations
    - **Read** — list observations or retrieve by ID
    - **Update** — modify a note field by ID
    - **Delete** — remove an observation by ID

External API Details

- **Geocoding endpoint**:

  `https://geocoding-api.open-meteo.com/v1/search`

  converts city name to latitude/longitude.
- **Weather endpoint**:

  `https://api.open-meteo.com/v1/forecast?current_weather=true`

  returns current temperature, windspeed, and timestamp.

Both are free, open, and require no API key.

API Endpoints Overview

Method	Endpoint	Description
POST	`/ingest?city=&country=`	Fetches live weather, saves it, returns record
GET	`/observations`	Retrieves all stored observations
GET	`/observations/{id}`	Retrieves a specific observation by ID

PUT	`/observations/{id}`	Updates the notes field of an observation
DELETE	`/observations/{id}`	Deletes an observation from DB

Sample Request & Response Scenarios

1. **Ingest Weather**

   `POST /ingest?city=Chicago&country=US`

   Response (200 OK)

   ```
   {
     "id": 1,
     "city": "Chicago",
     "country": "US",
     "latitude": 41.88,
     "longitude": -87.63,
     "temperature_c": 18.5,
     "windspeed_kmh": 12.3,
     "observation_time": "2025-09-04T12:00:00Z",
     "notes": null
   }
   ```

2. **List Observations**

   GET /observations

   Response


```
[
  {
    "id": 1,
    "city": "Chicago",
    "country": "US",
    "latitude": 41.88,
    "longitude": -87.63,
    "temperature_c": 18.5,
    "windspeed_kmh": 12.3,
    "observation_time": "2025-09-04T12:00:00Z",
    "notes": null
  },
  {
    "id": 2,
    "city": "Austin",
    "country": "US",
    "latitude": 30.27,
    "longitude": -97.74,
    "temperature_c": 26.1,
    "windspeed_kmh": 5.4,
    "observation_time": "2025-09-04T11:00:00Z",
    "notes": null
  }
]
```

3. **Add Note to an Observation**

```
PUT /observations/1
Content-Type: application/json
{
  "notes": "Perfect weather for a campus walk!"
}
```

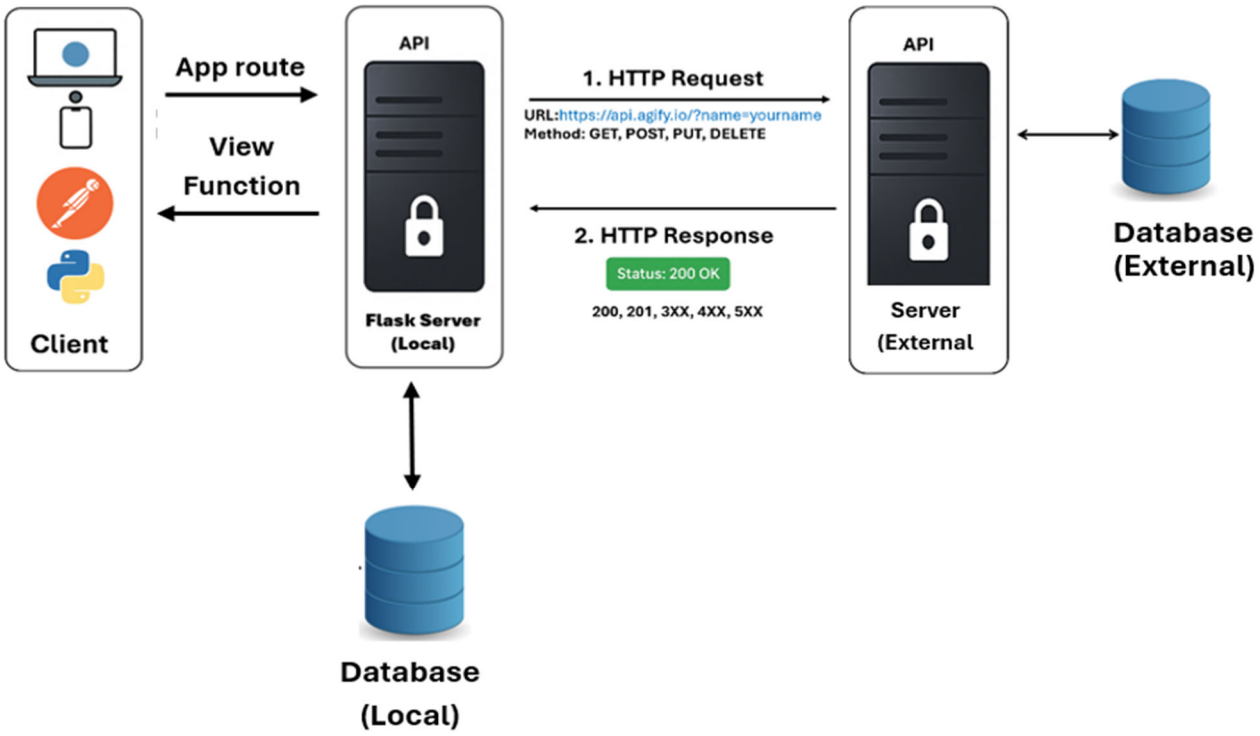
Response

```
{
  "id": 1,
  "notes": "Perfect weather for a campus walk!"
}
```

4. **Delete Observation**

```
DELETE /observations/1
```

Response

```
{
  "deleted": 1
}
```

Project Architecture